

CSC 470 — SOFTWARE ENGINEERING LAB

AI Resume Analyzer Platform

Activity Diagram • Swin Lane Diagram

Ehetisum Sharif

23103380

C# API Development & AI Integration

Nishat Tasnim

23103388

Database & Document Processing

Md. Sajid

23103399

Frontend UI & State Management

Submitted to: Avijit Biswas, Lecturer, Dept. of CSE, IUBAT School of CSE | Section D | Summer 2026 | 12/08/2026

Five functional modules

1

Authentication

Registration, login, role based session handling

2

Resume upload

File validation, preview and text extraction

3

AI scoring engine

ATS match score, categorised feedback, keywords

4

User dashboard

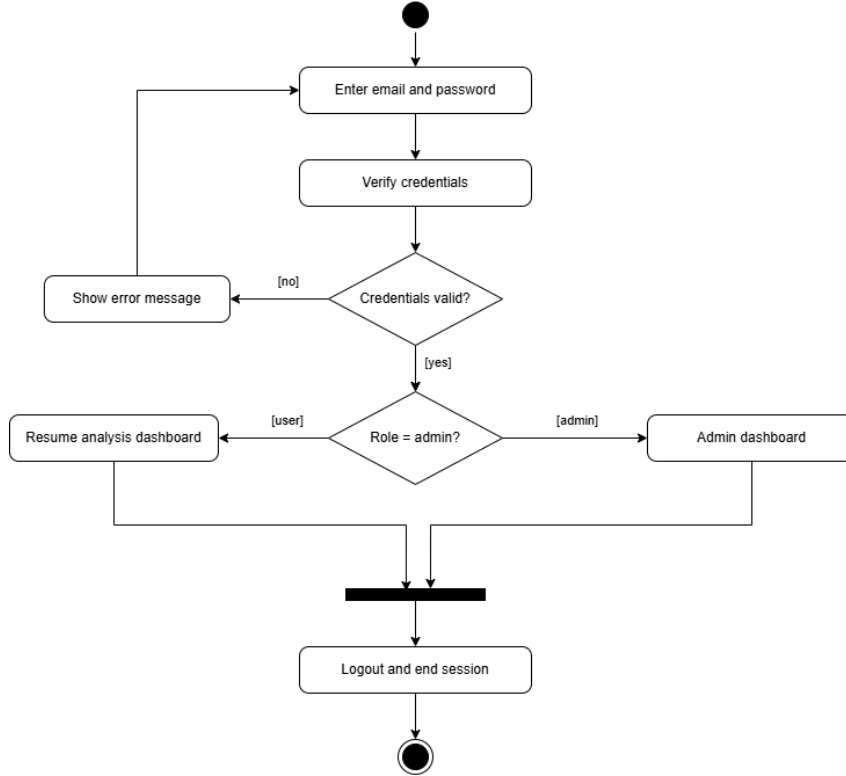
History, score over time chart, comparison view

5

Admin panel

Statistics, user management, system logs

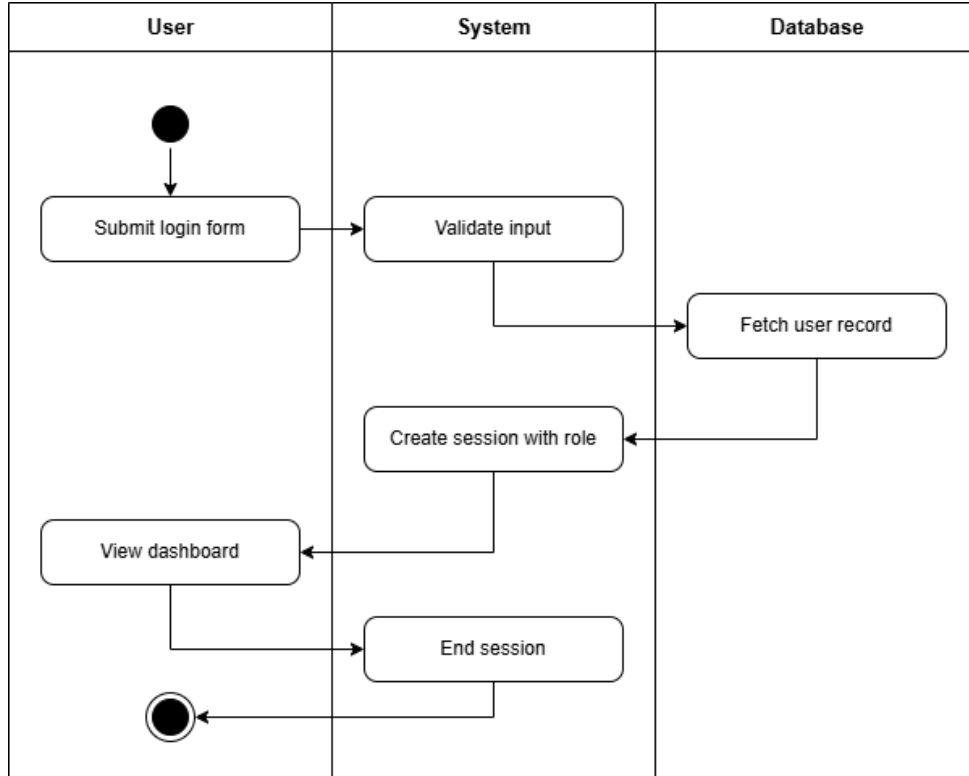
Authentication and session management



Key points

- Wrong password? The arrow goes back up to the login box.
- The second diamond checks the role: admin or normal user
- Both paths join again at logout, so there is only one end point

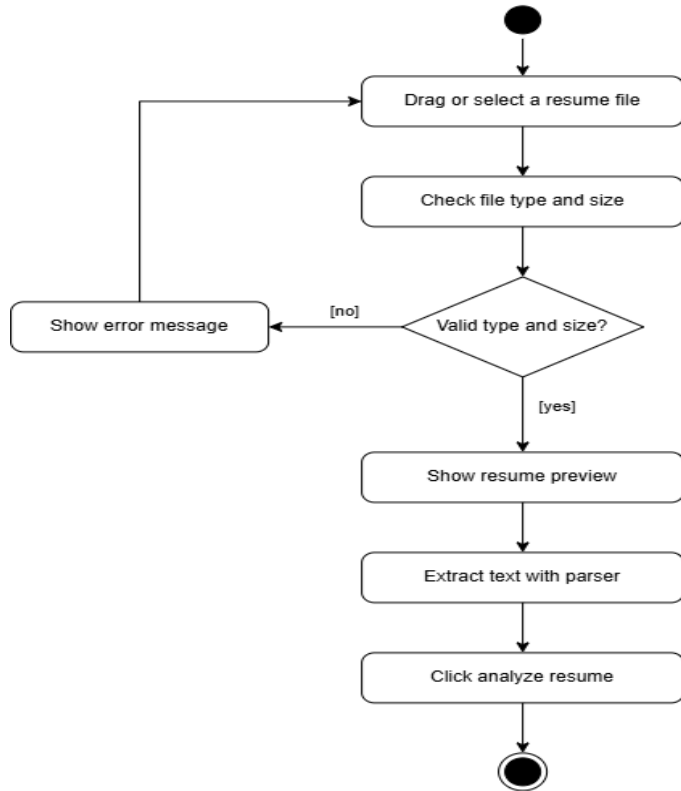
Authentication - who does what



Key points

- Three lanes: user, system, database
- The user only submits the form and then sees the dashboard
- The database is used one time, to find the user record
- Making and ending the session is the system's job

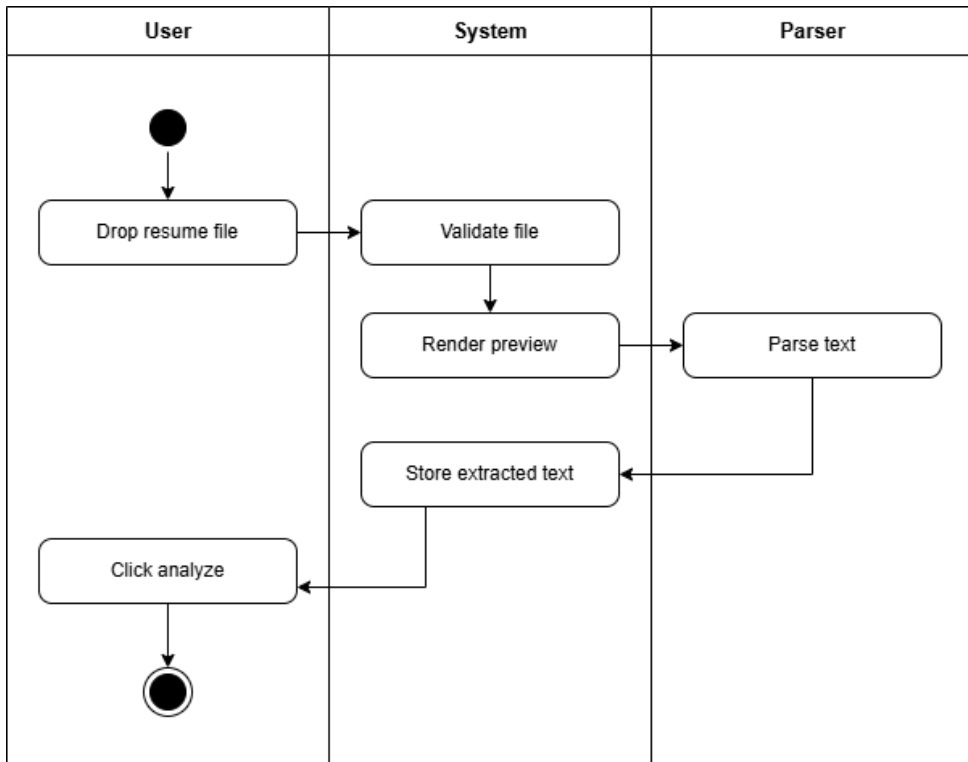
Resume upload and processing



Key points

- The diamond checks two things: file type and file size.
- Wrong file? Show the error and go back to upload
- The preview loads without refreshing the page.
- Text extraction is a separate box because an outside tool does it

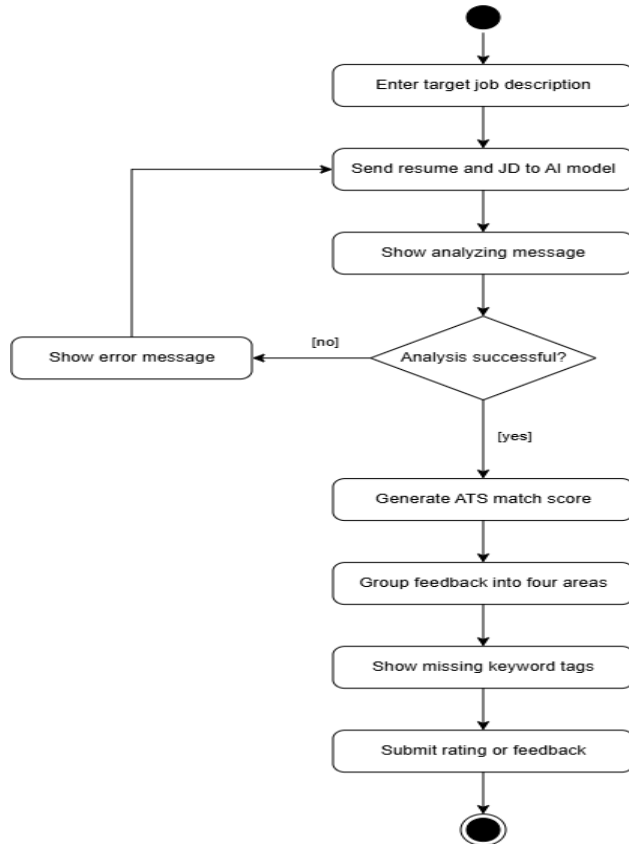
Resume upload - who does what



Key points

- The parser gets its own lane because it is not our code
- File checking stays with the system, not with the browser
- The last system step saves the text, ready for the AI module

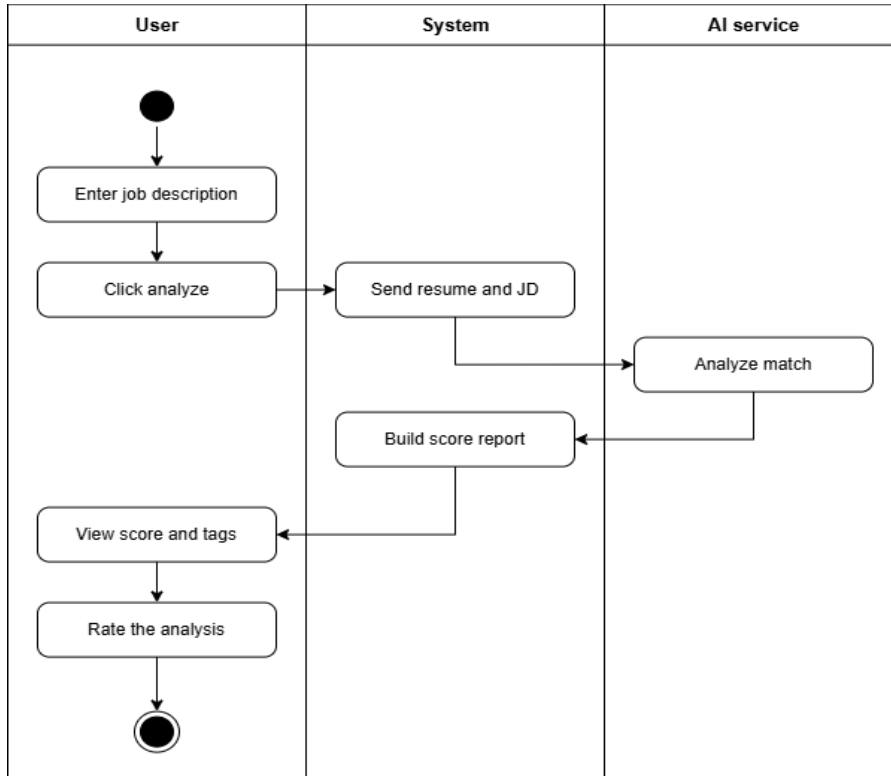
AI scoring and feedback engine



Key points

- The loading message is drawn as a real step.
- Rating is only possible after the analysis works.
- If the AI fails, we show an error and try the request again
- Score, four feedback types and keywords come from one AI reply

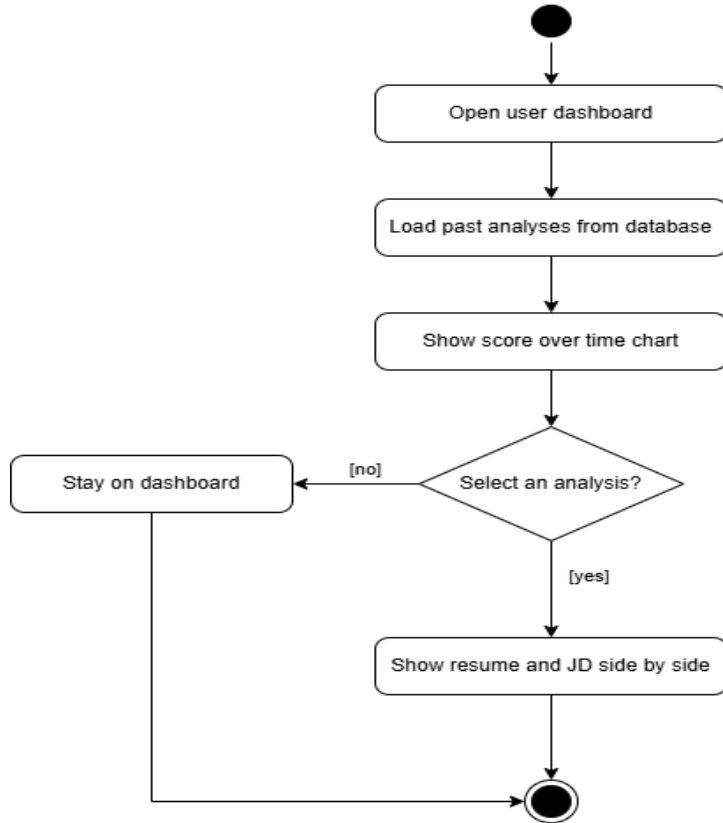
AI scoring - who does what



Key points

- The AI model is an outside service, so it gets its own lane
- Our system sends the request and builds the report
- The user only starts the work and reads the result

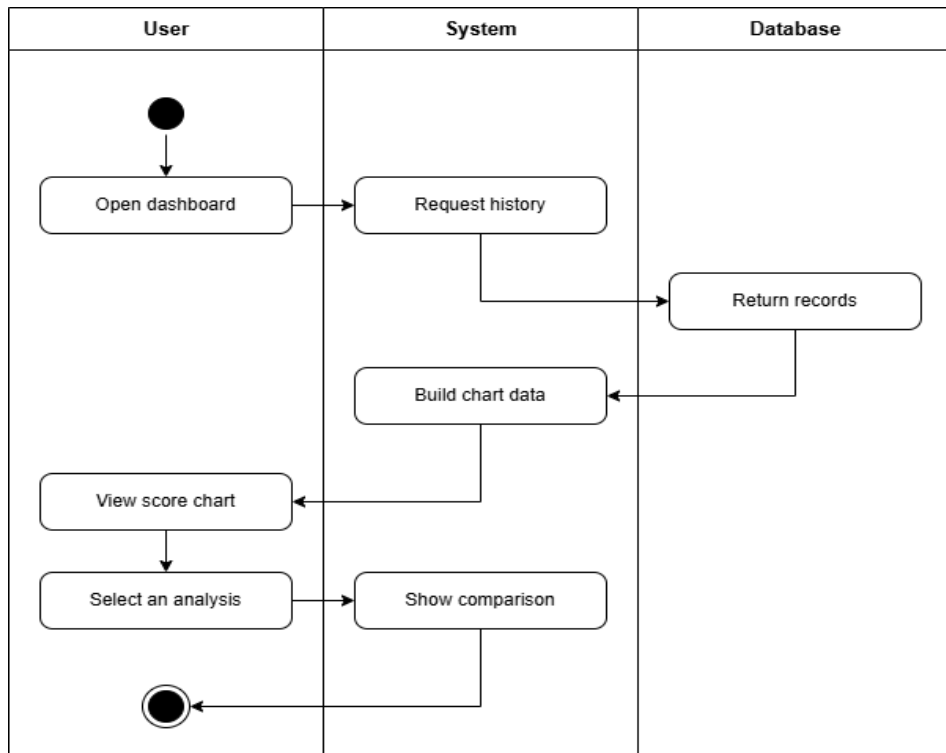
User dashboard and analytics



Key points

- History and the score chart load together when the page opens
- This diamond is a choice, not an error
- The user may compare an old analysis, or may simply stop here
- Both paths end at the same point

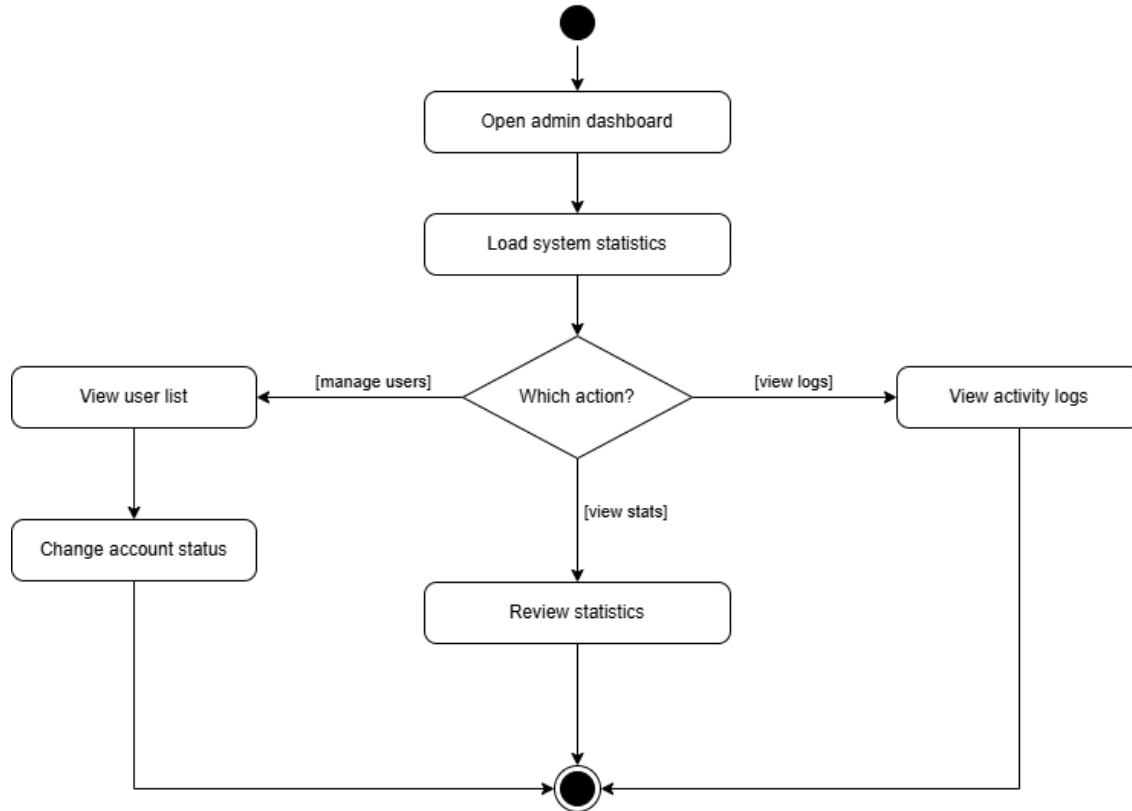
Dashboard - who does what



Key points

- The database only gives back raw records
- Turning those records into chart data is the system's job
- This keeps the layers separate, which NFR-5 asks for

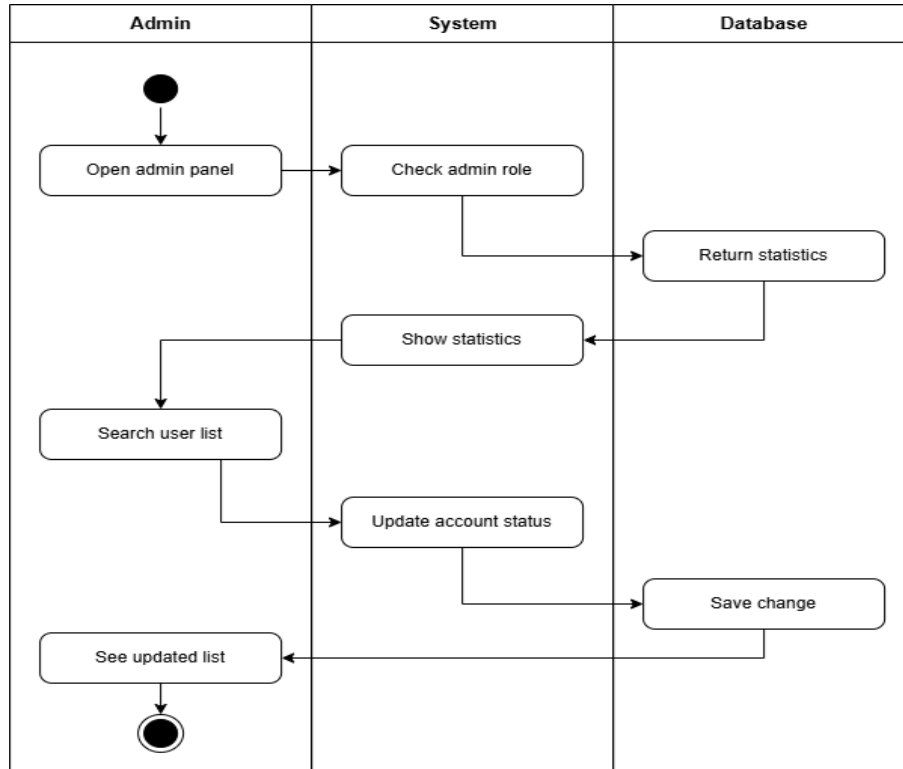
Admin panel management



Key points

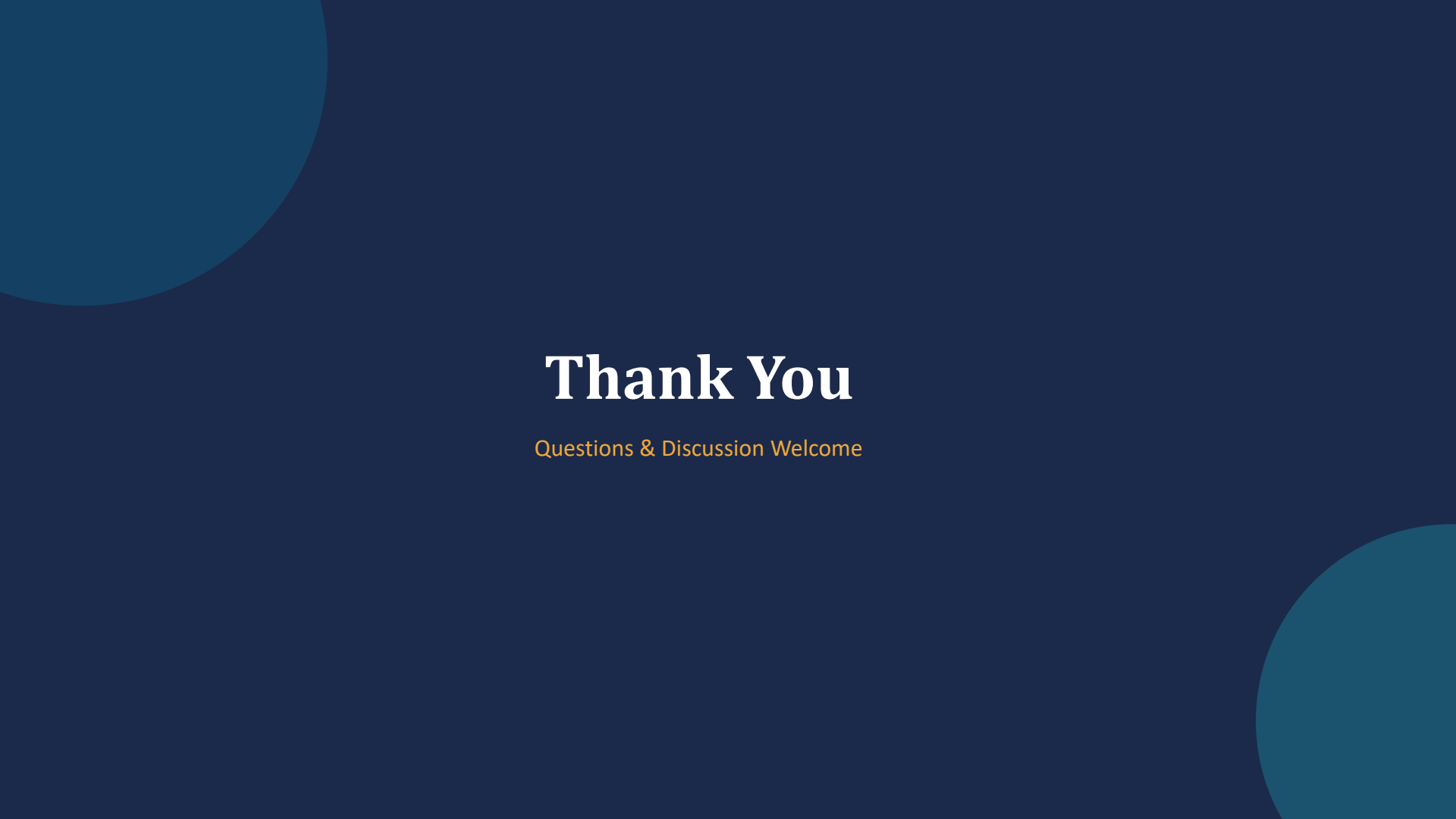
- One diamond splits into the three admin jobs
- Change status comes after the user list, because you must find the user first
- All three paths end at the same point

Admin panel - who does what



Key points

- The role check is the first system step.
- The database lane appears twice: one read, one write
- The admin never writes to the database directly, the system does it



Thank You

Questions & Discussion Welcome